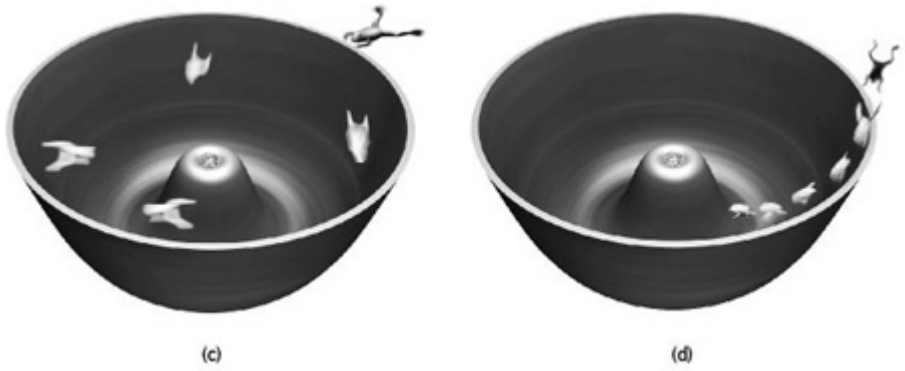


Figure 9.1 **(**c**)** As in (a), but with a hot bowl of a different shape. **(**d**)** As in (b), but now when the bowl cools, the frog slides down to the valley, which is some distance from the bowl's center (where the worms are located).

But here's the thing. As with the frog example, there's a possibility of two qualitatively different outcomes. If the shape of the field's energy bowl—its so-called *potential energy*—is similar to that in Figure 9.1a, the field's value throughout space will slide all the way down to zero, the bowl's center, just as the frog slides all the way down to the worm pile. However, if the field's potential energy looks like that in Figure 9.1c, the field's value will not make it all the way to zero, to the energy bowl's center. Instead, just as the frog will slide down to the valley, which is a *nonzero* distance from the worm pile, the field's value will also slide down to the valley—a nonzero distance from the bowl's center—and that means the field will have a nonzero value.[6] The latter behavior is characteristic of Higgs fields. As the universe cools, a Higgs field's value gets caught in the valley and never makes it to zero. And since what we're describing would happen uniformly throughout space, the universe would be permeated by a uniform and nonzero Higgs field—a Higgs ocean.

The reason this happens sheds light on the fundamental peculiarity of Higgs fields. As a region of space becomes ever cooler and emptier—as matter and radiation get ever more sparse—the energy in the region gets ever lower. Taking this to the limit, you know you've reached the emptiest a region of space can be when you've lowered its energy as far as possible. For ordinary fields suffusing a region of space, their energy contribution is lowest when their value has slid all the way down to the